Customers Like You

DS5020 Mini-Project: PCA or SVD for a real retail problem

Role	Name
Data Engineer	Achintya Shah
ML Engineer	Tanay Mihani

Git repository

https://github.com/AchintyaShah25/MiniProj_Linear_Algebra

Everything is reproducible from the repository: `python build_matrix.py` rebuilds the matrix from the raw file, then `ML_Engineer.ipynb` runs top to bottom. The README explains how to run it.

What is in this file

Pages	Contents
2	One-page memo to Sarah Okafor (J2)
3 to 11	Presentation slides (J3): slides 1 to 4 Achintya, 5 to 8 Tanay, 9 both

What is in the repository

Item	Where
README with row count (1,067,371), success targets, results, how to run, roles	`README.md`
One-command rebuild of the matrix from the raw file (DE6, R6)	`build_matrix.py`
Data profile, cleaning, time split, sparse matrix (DE1 to DE5)	`Data_Engineer.ipynb`
Cleaning log and matrix report (DE2, DE5)	`results/cleaning_log.csv`, `results/matrix_report.json`
Method choice, validation, recommendations, evaluation, PCA vs SVD table, customer map, purchase value check (ML1 to ML7)	`ML_Engineer.ipynb` (sections 2 to 16)
Figures and results (no customer IDs)	`figures/`, `results/`
Decision log (R1) and contribution log (C2)	`docs/`
Package versions and seeds (R6)	`requirements.txt`, seed 42

Headline results

Target	Result
Compressed data much smaller	7.3 MB of factors at k = 100, 22x smaller than the 160.6 MB dense matrix
Nightly job fast	About 1 second for the whole job
Beat the best-seller baseline	Hit rate at 10 of 40.9% vs 26.5% on 1,919 test customers
Customer groups marketing can name	6 named groups from their signature products

No customer IDs appear in this file, the memo, the slides or any committed file (R4).

Product suggestions and customer groups: what we built

To: Sarah Okafor, Head of E-Commerce | From: Data & ML Team (Achintya Shah, Tanay Mihani) | October 2026

In short. Our suggestions match what customers actually buy next about 1.5 times as often as showing our best sellers. The data the nightly job needs is 22 times smaller than the old table, and the whole job runs in under 2 seconds. For marketing, we found six customer groups, each with clear signature products.

Which method we chose, and why

We used a method called **truncated SVD**. It goes through two years of orders and finds 100 patterns in which products tend to be bought together. Every product and every customer is then described by 100 numbers instead of a row of 3,680 yes/no boxes. Products with similar numbers are bought by the same shops, which is exactly what a "customers who bought this also bought" box needs. We also tried the related method PCA on the same data. It gave almost the same results (40.2% against 40.9%) but needs an extra averaging step, so we kept SVD.

Does it work?

We held back the last two months of orders (9 Oct to 9 Dec 2011), built everything from the earlier orders only, and then checked our 10 suggestions for each of the 1,919 returning customers against what they really bought for the first time in those two months. For **41%** of them, at least one suggestion was something they went on to buy. Showing the 10 best sellers instead managed **27%**. On the White Hanging Heart T-Light Holder page, the box shows the red version of the same holder, the pink hanging heart candleholder, the hanging metal heart lantern, the cream heart card holder and the white metal lantern.

Size and speed

The compressed data is **7.3 MB**, against 161 MB for a full customer-by-product table. The whole nightly job (finding the patterns, building the box for all 3,680 products and picking 10 suggestions for each of 5,456 customers) takes **about a second** on a laptop.

What we lose by compressing

The 100 patterns keep about 36% of the detail in the order history. Most of what gets dropped is one-off orders that no other shop repeats, and those don't help suggestions anyway. What we do lose is the ability to suggest products without an established history: only products bought by at least 10 customers are included.

Customer groups for marketing

Group	Shops	What they buy
Heart and wicker home decor	1,321	Hanging and wicker hearts, zinc T-light holders. Almost all UK (98%)
Traditional toys and crafts	945	Vintage paint sets, modelling clay, dominoes, harmonicas
Funny metal signs	935	Novelty metal signs. Narrowest range of products
Regency tea and baking	830	Regency teacups, teapots and cake tins
Kids' party tableware	718	Skull, dinosaur and spotty paper plates and cake cases. Most international group
Jumbo bags and storage	707	Jumbo bags in many prints. Widest range of products

The groups overlap: they describe what a shop leans towards, not hard boxes. A shop can fit more than one campaign.

What could go wrong

- **New products** are never suggested until at least 10 customers have bought them. That was about 1 in 10 of the new purchases in our test. Until then they should fall back to the best sellers in their category.
- **New customers** have no order history, so they can only be shown best sellers.
- **Season.** We tested on the two months before Christmas 2011 only. Rebuilding the patterns every night, as planned, keeps them current.
- **Popular items** get a head start in the suggestions, so we should check the box doesn't show the same few products everywhere.
- **Missing data.** 23% of order lines had no customer number and couldn't be used.
- **Past orders only.** This was tested on history. The real proof is a live test on the site.

Next steps

(1) Run an A/B test of the box on part of the product pages for four weeks. (2) Add the fallbacks for new products and new customers. (3) Marketing pilots one campaign per customer group. **Privacy:** no customer numbers appear in this memo or our slides. They stay in one internal file inside the team.

Customers Like You

Product suggestions and customer groups for GiftLane

Achintya Shah, Data Engineer | Tanay Mihani, ML Engineer

All four targets from Sarah's email were met

22x

smaller

7.3 MB of factors instead of a 161 MB table

Target: under 16 MB

~1 s

nightly job

Fit, product page boxes and top 10 for every customer

Target: under 60 s

41%

hit rate at 10

vs 27% for best sellers, on 1,919 test customers

Target: beat best sellers

6

customer groups

Each named from the products it buys most

Target: 3 to 6 nameable groups

Cleaning kept 773,216 of 1.07 million invoice lines



72%

of the raw file kept

- Lines without a customer can't tell us anyone's taste, so they go first.
- Postage, fees and other non-product codes aren't purchases.
- Cancellations and items returned in full are removed inside each period, so the test months never affect training.

Split by time, then a matrix that is 98% empty

Training: Dec 2009 to 9 Oct 2011 | **Test: 2 months**

The model only ever sees training. The test months are the 'future' we check against: 1,919 returning customers and 49,214 products they bought for the first time.

5,456 x 3,680

customers x products (products bought by 10+ customers)

411,136

cells set to 1 (customer ever bought it)

97.95%

of cells are 0

4.96 MB

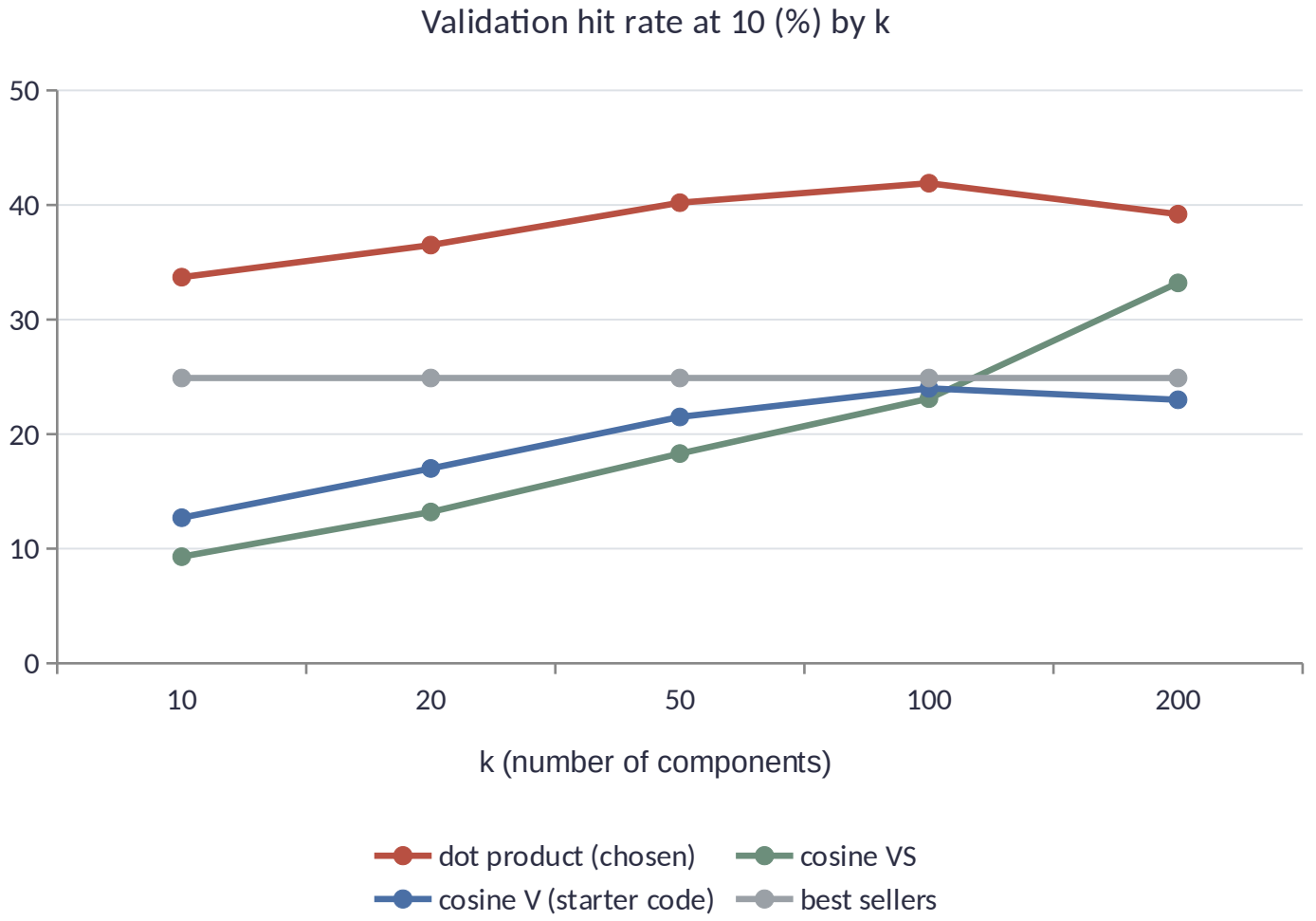
stored sparse (160.6 MB if stored dense)

SVD finds buying patterns; validation picked k = 100

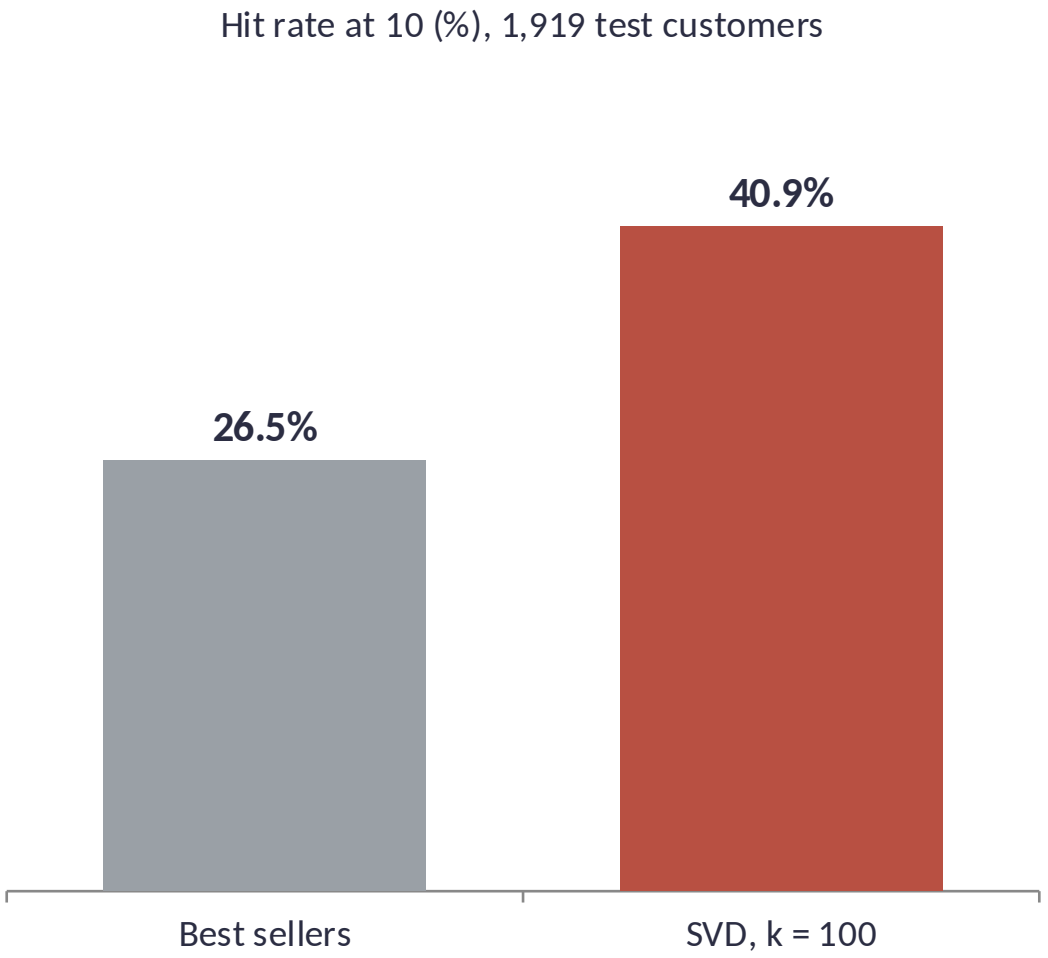
Why SVD
Works on the sparse matrix as it is, and gives product and customer factors in one fit.

How k was chosen
Learn from orders before 9 Aug 2011, check first-time buys up to 9 Oct. The test months stay untouched. Smallest k within 1 point of the best.

A surprise
The starter code's cosine scoring never beat best sellers. Scoring with plain dot products (the rank-k rebuild) did: 41.9% at k = 100.



On unseen purchases, suggestions beat best sellers 1.5 to 1



On the White Hanging Heart T-Light Holder page

- Red Hanging Heart T-Light Holder
- Candleholder Pink Hanging Heart
- Hanging Metal Heart Lantern
- Cream Heart Card Holder
- White Metal Lantern

Correct products per top 10: 0.66 vs 0.36 for best sellers.

Validation said 41.9%, test gave 40.9%: the validation split was a fair guide.

PCA and SVD came out almost the same on this matrix

Measure	PCA	Truncated SVD
Fit time	0.30 s	0.21 s
Peak memory during fit	16.6 MB	17.8 MB
Stored factors	7.34 MB	7.31 MB
Variance kept	36.2%	36.0%
Hit rate at 10 (test)	40.2%	40.9%
Component 1	basket size (corr 0.98)	basket size (corr 0.98)

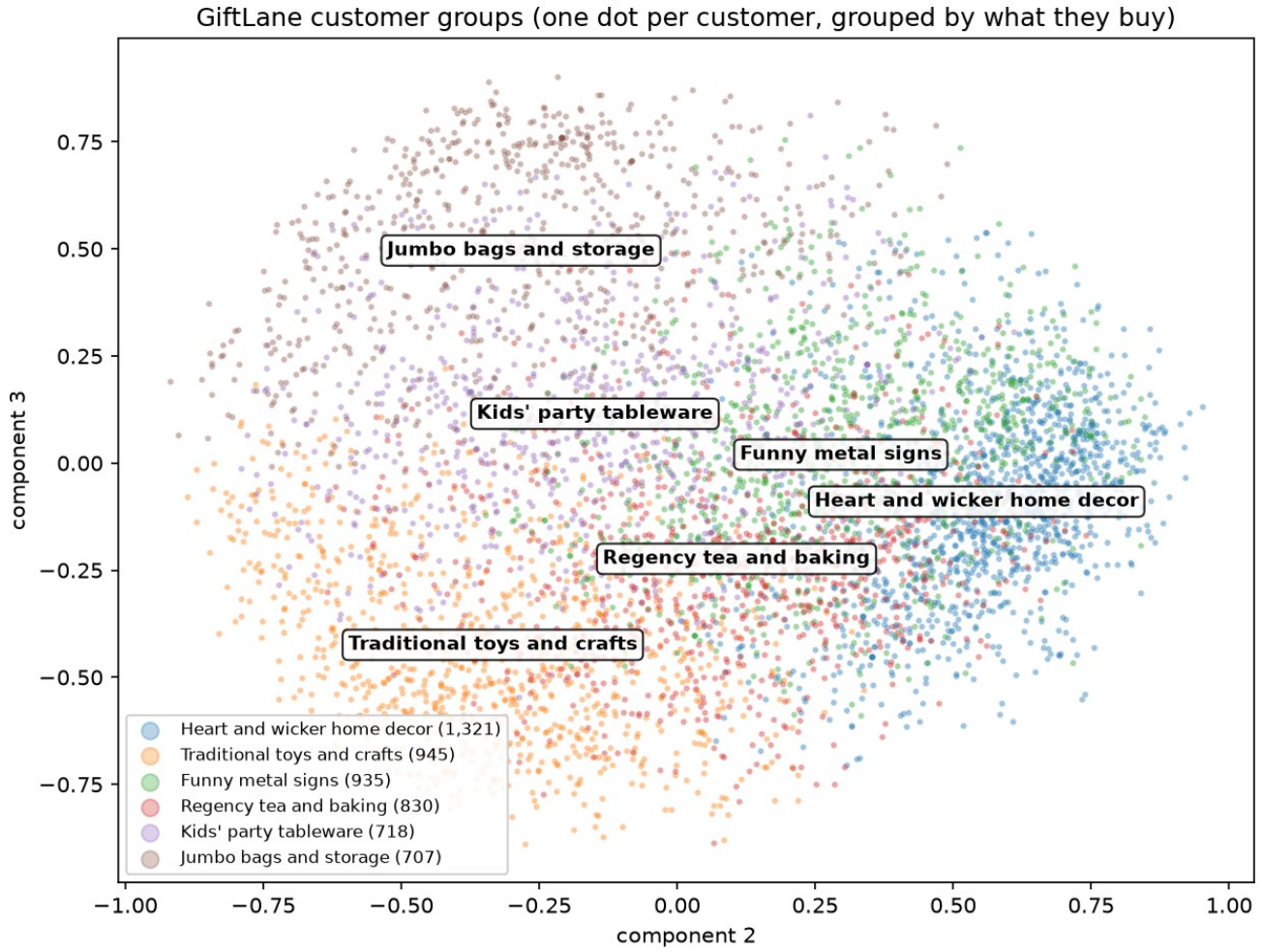
Why so close

Product averages are tiny: the median product was bought by 1.2% of shops. Subtracting them barely moves the matrix.

What we kept

SVD: same accuracy, a bit faster, and it works on the matrix exactly as built.

Six customer groups, named from what they buy



Heart and wicker home decor 1,321 shops
Hanging and wicker hearts; 98% UK

Traditional toys and crafts 945 shops
Paint sets, modelling clay, dominoes

Funny metal signs 935 shops
Novelty metal signs; narrowest range

Regency tea and baking 830 shops
Regency teacups, teapots, cake tins

Kids' party tableware 718 shops
Party paperware; most international

Jumbo bags and storage 707 shops
Jumbo bags; widest range

What could go wrong, and what's next

Risks

- New products (under 10 buyers) are never suggested: about 1 in 10 of the new purchases we tested
- New customers have no history, so they only get best sellers
- Tested on one season, the run-up to Christmas 2011
- Suggestions lean towards popular products
- Tested on past orders only

Next steps

- Four-week A/B test of the box on part of the product pages
- Fallbacks: category best sellers for new products and new customers
- Marketing pilots one campaign per group
- One-command rebuild (build_matrix.py) for the nightly run